

Homework 2 Mathematical Foundations of Reinforcement Learning

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Question 1

Part a)

We can simply write out the bellman equation for our non-terminal states. We start with $\mu_1(s) = 1$ which have the following bellman equations.

$$\begin{aligned}V_1 &= -1 + 0.9V_1 + 0.1V_2 \\V_2 &= -1 + 0.9V_2 + 0.1V_3 \\V_3 &= -1 + 0.9V_3\end{aligned}$$

We start from the bottom and solve

$$0.1V_3 = -1 \Rightarrow V_3 = -10$$

Substitute V_3

$$\begin{aligned}V_2 &= -1 + 0.9V_2 - 1 \\&\Rightarrow 0.1V_2 = -2 \\&\Rightarrow V_2 = -20\end{aligned}$$

Substitute V_2

$$\begin{aligned}V_1 &= -1 + 0.9V_1 + 2 \\&\Rightarrow 0.1V_1 = -3 \\&\Rightarrow V_1 = -30\end{aligned}$$

That gives us the value function

$$v_{\mu_1} = \begin{pmatrix} -30 \\ -20 \\ -10 \\ 0 \end{pmatrix}$$

For the second $\mu_2(s) = 2$ we have

$$V_1 = -2 + 0.5V_1 + 0.5V_2$$

$$V_2 = -2 + 0.5V_1 + 0.5V_3$$

$$V_3 = -2 + 0.5V_2$$

$$\begin{aligned} V_2 &= -2 + 0.5V_1 + 0.5(-2 + 0.5V_2) \\ &= -2 + 0.5V_1 - 1 + 0.25V_2 \\ &= -3 + 0.5V_1 + 0.25V_2 \\ \Rightarrow 0.75V_2 &= -3 + 0.5V_1 \end{aligned}$$

From

$$V_1 = -2 + 0.5V_1 + 0.5V_2,$$

we get

$$\begin{aligned} 0.5V_1 &= -2 + 0.5V_2 \\ \Rightarrow V_1 &= -4 + V_2. \end{aligned}$$

Substitute V_1 in the equation for V_2 :

$$\begin{aligned} 0.75V_2 &= -3 + 0.5(V_2 - 4) \\ &= -3 + 0.5V_2 - 2 \\ &= -5 + 0.5V_2 \\ \Rightarrow 0.25V_2 &= -5 \\ \Rightarrow V_2 &= -20. \end{aligned}$$

Then

$$\begin{aligned} V_1 &= V_2 - 4 = -24, \\ V_3 &= -2 + 0.5V_2 = -12. \end{aligned}$$

Therefore,

$$V_{\mu_2} = \begin{pmatrix} -24 \\ -20 \\ -12 \\ 0 \end{pmatrix}$$

Part b)

We perform policy iteration until we converge to the optimal policy. We use the μ_1 policy as a start. We use the optimal Bellman operator

$$V^*(s) = \max_{a \in A} \left[r(s, a) + \sum_{s' \in S} P(s' | s, a) V^*(s') \right]$$

So we start with policy

$$\pi^0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

For state 1, then

$$Q(1, 1) = -1 - 0.9 \cdot 30 - 0.1 \cdot 20 = -30$$

$$Q(1, 2) = -2 - 0.5 \cdot 30 - 0.5 \cdot 20 = -27$$

So the optimal a is 2.

$$Q(2, 1) = -1 - 0.9 \cdot 20 - 0.1 \cdot 10 = -20$$

$$Q(2, 2) = -2 - 0.5 \cdot 30 - 0.5 \cdot 10 = -22$$

So the optimal a is 1.

$$Q(3, 1) = -1 - 0.9 \cdot 10 = -10$$

$$Q(3, 2) = -2 - 0.5 \cdot 20 = -12$$

So the optimal a is 1.

The new policy is

$$\pi^1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

Then we need to solve the values of V^1

$$V_1 = -2 + 0.5V_1 + 0.5V_2$$

$$V_2 = -1 + 0.9V_2 + 0.1V_3$$

$$V_3 = -1 + 0.9V_3$$

$$\begin{aligned}
V_3 &= 10 \\
0.1 \cdot V_2 &= -2 \Rightarrow V_2 = -20 \\
V_1 &= -4 - 20 \Rightarrow V_1 = -24
\end{aligned}$$

with

$$V^1(s) = \begin{pmatrix} -24 \\ -20 \\ -10 \end{pmatrix}$$

We evaluate

$$\begin{aligned}
Q(1, 1) &= -1 - 0.9 \cdot 24 - 0.1 \cdot 20 = -1 - 21.6 - 2 = -24.6 \\
Q(1, 2) &= -2 - 0.5 \cdot 24 - 0.5 \cdot 20 = -2 - 12 - 10 = -24
\end{aligned}$$

so the optimal a is still 2.

$$\begin{aligned}
Q(2, 1) &= -1 - 0.9 \cdot 20 - 0.1 \cdot 10 = -20 \\
Q(2, 2) &= -2 - 0.5 \cdot 24 - 0.5 \cdot 10 = -19
\end{aligned}$$

so flip action to 2.

$$\begin{aligned}
Q(3, 1) &= -1 - 0.9 \cdot 10 = -10 \\
Q(3, 2) &= -2 - 0.5 \cdot 20 = -12
\end{aligned}$$

a stays as 1.

So the new policy is

$$\pi^2 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

We solve $V^2(s)$

$$\begin{aligned}
V_3 &= 10 \\
V_2 &= -2 + 0.5V_1 - 5 \Rightarrow V_2 = 0.5V_1 - 7 \\
\Rightarrow V_2 + 7 &= 0.5V_1 \Rightarrow 2V_2 + 14 = V_1 \Rightarrow V_2 = 0.5V_1 - 7
\end{aligned}$$

$$\begin{aligned}
0.5V_1 &= -2 + 0.5(0.5V_1 - 7) \\
&\Rightarrow -4 + 0.5V_1 - 7 \\
&\Rightarrow V_1 = -22
\end{aligned}$$

$$V_2 = -0.5 \cdot 22 - 7 = -18$$

Therefore

$$V^2(s) = \begin{pmatrix} -22 \\ -18 \\ -10 \end{pmatrix}$$

Iterate again

$$Q(1, 1) = -1 - 0.9 \cdot 22 - 0.1 \cdot 18 = -1 - 19.6 - 1.8 = -24.6$$

$$Q(1, 2) = -2 - 0.5 \cdot 22 - 0.5 \cdot 18 = -2 - 11 - 9 = -22$$

Keep action a as 2.

$$Q(2, 1) = -1 - 0.9 \cdot 18 - 0.1 \cdot 10 = -18.2$$

$$Q(2, 2) = -2 - 0.5 \cdot 22 - 0.5 \cdot 10 = -18$$

a stays 2.

$$Q(3, 1) = -1 - 0.9 \cdot 10 = -10$$

$$Q(3, 2) = -2 - 0.5 \cdot 18 = -11$$

a stays 1.

So the policy did not change this iteration which means we now have arrived at the optimal policy

$$\pi^* = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

Part c)

In part 2 and section 2.2 in neuro-dynamic programming the authors claim that if every stationary policy is proper (i.e., they eventually terminate) then the optimal Bellman operator is contractive on a weighted max-norm. We will assume this is true without proof, but we can assume it because there is always a small chance of existing the chain so after a sufficient amount of iterations the probability of terminating should approach 1.

Given this, the author shows that we can derive the weights for such a norm by finding the solution for the SPP of the non-terminal states where the cost is -1. Then we can find the weights for the norm by taking the negative of the solution $\mathcal{E} = -V$ of this problem. This is essentially the worst case for the number of non-terminal states we visit if we start in some state.

They then also derive that we can find a constant

$$\beta = \max_s \frac{\mathcal{E}_s - 1}{\mathcal{E}_s}$$

for which the weighted norm is contractive.

We can solve this new problem using policy iteration.

$$\begin{aligned}\mathcal{E}_1 &= 1 + \max\{0.9\mathcal{E}_1 + 0.1\mathcal{E}_2, 0.5\mathcal{E}_1 + 0.5\mathcal{E}_2\} \\ \mathcal{E}_2 &= 1 + \max\{0.9\mathcal{E}_2 + 0.1\mathcal{E}_3, 0.5\mathcal{E}_1 + 0.5\mathcal{E}_3\} \\ \mathcal{E}_3 &= 1 + \max\{0.9\mathcal{E}_3, 0.5\mathcal{E}_2\}\end{aligned}$$

We start again with the fixed policy $\mu_1 = 1$ for all states. Then, the expected value is

$$V = \begin{pmatrix} 30 \\ 20 \\ 10 \end{pmatrix}$$

So, we compute for all states the best action.

$$\begin{aligned}\max_a \{0.9 \cdot 30 + 0.1 \cdot 20, 0.5 \cdot 30 + 0.5 \cdot 20\} &\Rightarrow 1 \\ \max_a \{0.9 \cdot 20 + 0.1 \cdot 10, 0.5 \cdot 30 + 0.5 \cdot 10\} &\Rightarrow 2 \\ \max_a \{0.9 \cdot 10, 0.5 \cdot 20\} &\Rightarrow 2\end{aligned}$$

which gives us actions $(1, 2, 2)$.

And then we evaluate this new policy

$$\begin{aligned}\mathcal{E}_1 &= 1 + 0.9\mathcal{E}_1 + 0.1\mathcal{E}_2 \\ \mathcal{E}_2 &= 1 + 0.5\mathcal{E}_1 + 0.5\mathcal{E}_3 \\ \mathcal{E}_3 &= 1 + 0.5\mathcal{E}_2\end{aligned}$$

$$\begin{aligned}\mathcal{E}_2 &= 1 + 0.5\mathcal{E}_1 + 0.5(1 + 0.5\mathcal{E}_2) \\ \Rightarrow \mathcal{E}_2 &= 1 + 0.5\mathcal{E}_1 + 0.5 + 0.25\mathcal{E}_2 \\ \Rightarrow 0.75\mathcal{E}_2 &= 1 + 0.5\mathcal{E}_1 + 0.5 \\ \Rightarrow 0.75\mathcal{E}_2 - 1.5 &= 0.5\mathcal{E}_1 \\ \Rightarrow 1.5\mathcal{E}_2 - 3 &= \mathcal{E}_1 \\ \Rightarrow \mathcal{E}_2 &= \frac{2}{3}\mathcal{E}_1 + 2\end{aligned}$$

$$\begin{aligned}
\mathcal{E}_1 &= 1 + 0.9\mathcal{E}_1 + 0.1\left(\frac{2}{3}\mathcal{E}_1 + 2\right) \\
\Rightarrow \mathcal{E}_1 &= 1 + \frac{27}{30}\mathcal{E}_1 + \frac{2}{30}\mathcal{E}_1 + 0.2 \\
&= \mathcal{E}_1 = \frac{29}{30}\mathcal{E}_1 + 1.2 \\
\Rightarrow \frac{1}{30}\mathcal{E}_1 &= +1.2 \Rightarrow \mathcal{E}_1 = 36
\end{aligned}$$

$$\begin{aligned}
\mathcal{E}_2 &= \frac{2}{3}\mathcal{E}_1 + 2 = 26 \\
\mathcal{E}_3 &= 1 + 0.5\mathcal{E}_2 = 14
\end{aligned}$$

Evaluate the policy again

$$\begin{aligned}
\max_a \{0.9 \cdot 36 + 0.1 \cdot 26, 0.5 \cdot 36 + 0.5 \cdot 26\} &\Rightarrow 1 \\
\max_a \{0.9 \cdot 26 + 0.1 \cdot 14, 0.5 \cdot 36 + 0.5 \cdot 14\} &\Rightarrow 2 \\
\max_a \{0.9 \cdot 14, 0.5 \cdot 26\} &\rightarrow 2
\end{aligned}$$

and see that we have converged.

Then, our current value of $\mathcal{E}$ is the weight vector for which the operator is contractive and where

$$\beta = \max_i \frac{\mathcal{E}_i - 1}{\mathcal{E}_i} = \frac{35}{36}$$

Thus, the Bellman operator of the current problem is β -contractive on norm $\|x\|_w = \max_i \frac{|x_i|}{w_i}$ where

$$w = \begin{pmatrix} 36 \\ 26 \\ 14 \end{pmatrix}$$

and

$$\beta = \frac{35}{36}$$

Question 2

0.1 Part a)

Recall that the row i in a transition matrix represents the probability of moving to states $s \in S$ from state s_i , such that the $P(i, j)$ is the probability of moving from s_i to s_j .

We know that s_0 is the terminal state, from which we can never escape. Thus, the probability of $P(0,0)$ should be 1 and all other $P(i,:)$ should be zero, which describes the first row.

We are also informed that all other states have a non-zero probability of moving to the terminal state. Meaning that we have a non-zero column vector $P_1 = P(:,0)$.

For the remaining $P_2 = P(1:,1:)$ we know that it must be a matrix where each row will always sum to less than 1.

Part b)

we know from the Invertible Matrix Theorem that a matrix is singular, if and only if it has 0 as an eigenvalue. Thus we can prove 0 is not an eigenvalue of $(I - P_2)$, then it is not singular.

If we think about this, it means that if there is a non-zero vector w for which the dot product of each row in P_2 is 0.

If we now consider such a vector

$$\begin{aligned} (I - P_2)w &= \mathbf{0} \\ \Rightarrow Iw - P_2w &= \mathbf{0} \\ \Rightarrow w - P_2w &= \mathbf{0} \\ \Rightarrow P_2w &= w. \end{aligned}$$

Intuitively, this seems unlikely since each row in P_2 sums to less than one which should reduce the spectral radius of w so we should look at how the multiplication affects the max value of w . Suppose that $\max_j |w_j| = M$.

Look at the row where $|w_i| = M$. If we try and compute the value

$$|w_i| = |(P_2w)_i| = \sum_j p_{ij}|w_j| \leq \sum_j p_{ij}M = M \sum_j p_{ij}$$

But the row sum of any row in P_2 is always less than one, so $|(P_2w)_i| < M$ but if $P_2w = w$ then $|(P_2w)_i| = M$, which is a contradiction.

Thus, 0 cannot be a eigenvalue and therefore the matrix is non-singular.

Part c)

The terminal state $s = 0$ will by definition have reward zero, and since it has not transitions $v(0)$ is also 0. Furthermore, it means that state $s = 0$ will contribute 0 to each row when computing Pv , which means we can simply ignore it in our calculation, meaning we can also discard P_1 from our calculations.

Therefore, if we ignore $s = 0$, and focus on the remaining value vector $\hat{v}$

$$\begin{aligned}
\hat{v} &= \hat{r} + [P_1 \quad P_2] v \\
&= \hat{r} + P_2 \hat{v} \\
\Rightarrow \hat{v} - P_2 \hat{v} &= \hat{r} \\
\Rightarrow (I - P_2) \hat{v} &= \hat{r} \\
\Rightarrow \hat{v} &= (I - P_2)^{-1} \hat{r}.
\end{aligned}$$

Question 3

Part a)

We already showed this in Part c) of question since we can formulate the standard Bellman operator

$$\hat{v} = \hat{r} + P_2 \hat{v}$$

From this standard Bellman equation formulation we can then create the dynamic programming operator

$$\hat{v}^{k+1} = \hat{r} + P_2 \hat{v}^k$$

Part b)

I think the question is actually asking how many states we will visit if we start at s , because otherwise I don't see how that equation would make sense. So I will proceed with that understanding.

We can express that formulation as

$$N(s) = \mathbf{E} \left[\sum_{k=0}^{\tau-1} 1 \mid s^k = s \right].$$

where τ is the time it takes to reach terminal state $s = 0$.

This is essentially just a reformulation of our standard MRP or SPP where the reward is 1, and should have a solution of the policy is proper, i.e., the probability of reaching the terminal state moves to 1 as times goes on.

So if the reward is 1 for all non-terminal states then we can write the above formulation to the standard Bellman operator for a single state.

$$N(s) = 1 = \sum_{s' \in S_+} P(s, s') N(s')$$

Question 4

All norms expressions here uses the ∞ -norm.

We want to compute upper bound of the norm on the difference between $\hat{v} - v$. We simplify the expression as such

$$\hat{v} - v = (I - \gamma P)^{-1}(\hat{R} - R)$$

We will then use the inequality in the notes $\|Ax\| \leq \|A\|\|x\|$ to

We use the question hint and find an upper bound on the matrix inverse. We know that

$$\|(I + B)^{-1}\| \leq \frac{1}{1 - \|B\|}$$

if $\|B\| < 1$ If we then consider $B = -\gamma P$. We know that $\|B\| < 1$ since P is a stochastic matrix and γ is less than 1. Therefore, we express the upper bound of the

$$\|(I - \gamma P)^{-1}\| \leq \frac{1}{1 - \|-\gamma P\|} = \frac{1}{1 - \gamma}$$

We know that $\|\hat{R} - R\| \leq \epsilon$, thus we can combine the upper bound of the two expressions and the fact that $\|Ax\| \leq \|A\|\|x\|$ to

$$\begin{aligned} \|\hat{v} - v\| &\leq \|(I - \gamma P)^{-1}\| \|\hat{R} - R\| \\ &\leq \frac{1}{1 - \gamma} \gamma = \frac{\gamma}{1 - \gamma} \end{aligned}$$

Part b)

We know that the bellman operator converges on the true v

$$\lim_{k \rightarrow \infty} B(\hat{v}_k) \rightarrow v$$

So if we apply the Bellman operator we have an error bound ϵ . But if we apply it continuously, it will be bound the following series where it gets discounted at every step, creating a standard geometric series.

$$\begin{aligned} \|\hat{v} - v\| &= \|\hat{v} - B^\infty(\hat{v})\| \\ &\leq \epsilon + \gamma\epsilon + \gamma^2\epsilon \dots \\ &= \frac{\epsilon}{1 - \gamma} \end{aligned}$$

Part c)

Given that we compute the approxiamte value like this $\hat{v} - v = (I - \gamma\hat{P})^{-1}R$

We can just construct these two using the bellman operator instead.

Thus,

$$\begin{aligned}
 \hat{v} &= R + \gamma\hat{P}\hat{v} \\
 v &= R + \gamma\hat{P}v \\
 \hat{v} - v &= R + \gamma\hat{P}\hat{v} - (R + \gamma Pv) \\
 &= \gamma\hat{P}\hat{v} - \gamma Pv \\
 &= \gamma((P - P_{\text{diff}})\hat{v} - Pv) \\
 &= \gamma P(\hat{v} - v) - \gamma P_{\text{diff}}\hat{v} \\
 \Rightarrow (I - \gamma P)(\hat{v} - v) &= -\gamma P_{\text{diff}}\hat{v} \\
 \Rightarrow \hat{v} - v &= -\gamma(I - \gamma P)^{-1}P_{\text{diff}}\hat{v}.
 \end{aligned}$$

where $P_{\text{diff}} = P - \hat{P}$

From here we can use the same inequalities we used as in Part a. We know from the question that $\|P_{\text{diff}}\| \leq \epsilon$

$$\|\hat{v} - v\| = \frac{\gamma\epsilon}{1 - \gamma}\|\hat{v}\|$$

The thing is I cannot find a lower bound that removes the dependence on $\hat{v}$, which is just a dependency on R .

I will just show for fun. Assume that $\|R\| \leq R_{\text{max}}$

$$\begin{aligned}
 \hat{v} &= R + P\hat{v} \\
 \Rightarrow \|\hat{v}\| &\leq \|R\| + \gamma\|P\|\|\hat{v}\| = \|R\| + \gamma\|\hat{v}\| \\
 &\Rightarrow (1 - \gamma)\|\hat{v}\| = R_{\text{max}} \\
 &\Rightarrow \|\hat{v}\| \leq \frac{R_{\text{max}}}{1 - \gamma}
 \end{aligned}$$

So, the lowest bound I can find is

$$\|\hat{v} - v\| = \frac{\gamma\epsilon R_{\text{max}}}{(1 - \gamma)^2}$$

Question 5

Solving the linear system involves a few matrix operations. A matrix addition, but since it is the singular matrix it can be made in linear time, then a scalar multiplication of γ . Then a matrix inversion and then a matrix-vector multiplication

The time-complexity of finding the inverse of an size $N \times N$ matrix is $O(N^3)$, while all other operations are either linear or $O(N^2)$ so solving the linear system of an MRP with a statespace of size N is $O(N^3)$.

Looking at the dynamic programming approach, applying the bellman operator a single if for a similar state space space instead consist of a is a single matrix multiplication, a scalar multiplication and addition for each state which is $O(N^2 + N + N) = O(N^2)$. Therefore, as long as it takes less than N steps to converge then the dynamic programming approach is generally more efficient.

Question 6

The ∞ -norm on the Q -table is the single entry with the highest absolute value. We can interpretate what we are asked to prove is that the largest update between shrinks by a factor of at least γ .

$$\|Q_1 - Q_2\|_\infty = \max_{s,a} |Q_1(s,a) - Q_2(s,a)|$$

$$(TQ_1)(s,a) = r(s,a) + \gamma \mathbf{E}_{s'} [\max_{a'} Q_1(s',a')]$$

Check for some random pair of Q -tables

$$(TQ_1)(s,a) - (TQ_2)(s,a) = \gamma \mathbf{E}_{s'} [\max_{a'} Q_1(s',a') - \max_{a'} Q_2(s',a')]$$

We know that

$$|\mathbf{E}[X]| \leq \mathbf{E}[|X|]$$

$$|(TQ_1)(s,a) - (TQ_2)(s,a)| \leq \gamma \mathbf{E}_{s'} [|\max_{a'} Q_1(s',a') - \max_{a'} Q_2(s',a')|]$$

Of course, taking the maximum value of the difference is larger or equal to the difference between two max values.

$$|\max_{a'} Q_1(s',a') - \max_{a'} Q_2(s',a')| \leq \max_{a'} |Q_1(s',a') - Q_2(s',a')|$$

Recall that the ∞ -norm is the max over all states and all actions, which is either larger or equal for the max with a fixed state s'

$$\max_{a'} |Q_1(s', a') - Q_2(s', a')| \leq \|Q_1 - Q_2\|_\infty$$

So we replace the expected value from before with this norm

$$|(TQ_1)(s, a) - (TQ_2)(s, a)| \leq \gamma \|Q_1 - Q_2\|_\infty$$

which means that it is γ -contractive on the ∞ -norm. From the fixed point theorem we then have that T has a fixed point Q^* which satisfies the Bellman optimality equation

$$Q^* = TQ^*$$

meaning that it is the optimal policy.

So as usual, if we apply the Bellman operator k times we have an upper error bound

$$\gamma^k \|Q_0 - Q^*\|_\infty$$

Say that we want an upper error bound of ϵ then we need to run iteratively for k steps where k is

$$\begin{aligned} \gamma^k \|Q_0 - Q^*\|_\infty &\leq \epsilon \\ \Rightarrow k \log \gamma + \log(\|Q_0 - Q^*\|_\infty) &\leq \log \epsilon \\ \Rightarrow \log(\|Q_0 - Q^*\|_\infty) - \log \epsilon &\leq k \log \gamma \\ \Rightarrow \log(\|Q_0 - Q^*\|_\infty / \epsilon) &\leq -k \log \gamma \\ \Rightarrow \frac{\log(\|Q_0 - Q^*\|_\infty / \epsilon)}{-\log \gamma} &\leq k \end{aligned}$$

Question 7

Part a)

This chain will just oscillate between the two states, so the rewards will cancel out as we continue to run and the gain will be 0.

Similarly, the bias series will start with either one of the rewards of their respective states and then oscillate between that initial reward and 0. If we take the expected value of the Cesaro limit we get $h(1) = \frac{1}{2}$ and $h(2) = \frac{1}{2}$

Part b)

Recall from the last homework that a transition matrix that is column stochastic have the uniform distribution as its stationary distribution. P is column stochastic, although not aperiodic, but it still holds that the stationary distribution is $\pi(s) = 0.5$.

Just checking we see that this holds $\pi = \pi P$ which it $0.5 = 0 \cdot 0.5 + 1 \cdot 0.5$ and $0.5 = 1 \cdot 0.5 + 0 \cdot 0.5$

Part c

We can again express the value as a geometric series. For state one we start with reward 1 and then the sign will continue to flip.

$$v_\lambda(1) = 1 - \lambda + \lambda^2 - \lambda^3 + \dots$$

We can split the sum into even and odd powers,

$$v_\lambda(1) = (1 + \lambda^2 + \lambda^4 + \dots) - (\lambda + \lambda^3 + \lambda^5 + \dots).$$

Both are geometric series with ratio λ^2 , so

$$v_\lambda(1) = \frac{1}{1 - \lambda^2} - \frac{\lambda}{1 - \lambda^2} = \frac{1 - \lambda}{1 - \lambda^2} = \frac{1}{1 + \lambda}.$$

For state 2, we have the same but the sign starts as negative.

$$v_\lambda(2) = -1 + \lambda - \lambda^2 + \lambda^3 - \dots$$

and therefore

$$v_\lambda(2) = -\frac{1}{1 - \lambda^2} + \frac{\lambda}{1 - \lambda^2} = \frac{\lambda - 1}{1 - \lambda^2} = -\frac{1}{1 + \lambda}.$$

We know from the lecture that the gain can be computed as the dot product of the stationary distribution and the value function. But since the two discounted values are each others negatives, and the distribution is uniform, the gain must be 0.

Part d)

So we can setup this linear system and try and find a solution for g

$$\begin{aligned} g + h_1 - h_2 &= 1 \\ g - h_1 + h_2 &= -1 \end{aligned}$$

Combining these two expressions gives us

$$\begin{aligned} g &= h_2 - h_1 + 1 \\ \Rightarrow (h_2 - h_1 + 1) - h_1 + h_2 &= -1 \\ \Rightarrow 2h_2 - 2h_1 + 1 &= -1 \\ \Rightarrow h_2 - h_1 &= -1 \\ \Rightarrow h_2 - h_1 + 1 &= 0 \end{aligned}$$

But note that we also have that $g = h_2 - h_1 + 1$ and we also see that $h_2 - h_1 + 1 = 0 = g$, which means that g must always be 0, regardless of the value h_1 and h_2 .

Question 8

Part a)

We start by showing that the diff between two L operators is just a matrix multiplication $L(w) - L(u) = R + Pw - R + Pu = P(w - u)$, which means we can express the following $\text{sp}(L(w) - L(u)) \leq \gamma \text{sp}(w - u)$ holds. This is very similar to the usual proof of convergence to a fixed point for a contractive function, so we deconstruct the problem in a similar way. However, parts of that proof would rely on h^* being a fixed point of L , but according to the definition

$$g^* \mathbf{1} + h^* = r + Ph^*$$

this is not true since have a gain residual $g^* \mathbf{1}$. However, recall from the lecture that the sp-norm ignores constants. So $\text{sp}(g^* \mathbf{1} + h^*) = \text{sp}(h^*)$ which we can utilize.

As a result, we then know that

$$\text{sp}(h_{k+1} - h^*) = \text{sp}(P(h_k - h^*))$$

meaning that applying the transition matrix updates the bias iteration but keeps the "fixed point" fixed under the sp-norm. We can then express step k of the difference between h_k and the true h^* under sp as the application of the transition matrix k times.

$$\text{sp}(h_k - h^*) = \text{sp}(P^k(h_0 - h^*))$$

and since we know that each step have an upper error bound of γ we can construct an upper bound as follows

$$\text{sp}(h_k - h^*) = \text{sp}(P^k(h_0 - h^*)) \leq \gamma^k \text{sp}(h_0 - h^*)$$

Part b)

Recall from the lectures that the sp-norm only measures the maximum distance between elements in the vector, and thus ignores constants. Therefore, any inequality that holds for any $\text{sp}(x)$ will also hold $\text{sp}(x + c\mathbf{1})$.

If we at the same time know that $\text{sp}(h - h^*) \leq \epsilon$, that means that the $\max(h - h^*) - \min(h - h^*) \leq \epsilon$ which can be interpreted as the difference between the largest difference and the smallest difference is ϵ .

If we express this difference as a vector $\hat{e}$ where

$$h - h^* = \hat{e} = e + c\mathbf{1}$$

where $\hat{e}$ is the difference between the biases, which then consists of e which is a zero centered error vector and a constant c that is the center of e ' of max and min values.

I.e,

$$c = \frac{\max(h - h^*) + \min(h - h^*)}{2}$$

then we know since $sp(h - h^*) \leq \epsilon$ then this must hold for the zero-center vector e

$$\begin{aligned}\min(e) &\geq -\epsilon/2 \\ \max(e) &\leq \epsilon/2\end{aligned}$$

Recall that the ∞ -norm is simply the maximum absolute value of the vector, which means that $\|e\|_\infty \leq \epsilon/2$.

Thus, we have show that

$$h = h^* + e + c\mathbf{1}$$

where $\|e\|_\infty \leq \epsilon/2$